Lecture Scribe and LaTeX

L20.pdf
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Generate a lecture scribe. Use only the provided context. Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples exactly as presented in the lecture materials. Do not introduce new examples or explanations. Organize the scribe clearly so it can be used for revision. Only include few important images. Its okay if it does not have all the images. Do not change any formats also. I do also need the latex code for this pdf do also generate that. Do not cut any written text or content in this pdf. I want the whole same sentences same as the sentences in pdf.I wont upload the pdf again please read this only. Do not cut anything from this PDF

Got it — you want a **faithful lecture scribe + full LaTeX code**, with **no content removed or reworded**, exactly as in the PDF.

Below is your **structured lecture scribe**, followed by **complete LaTeX code** that reproduces it.

📘 LECTURE SCRIBE (Exact Content Preservation)

CSE400 - Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple RVs

Dhaval Patel, PhD
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Ahmedabad University, Ahmedabad, Gujarat- India

March 19, 2025

Outline

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

Case Study: "Smart City Traffic + Weather System"

Imagine you are designing an intelligent system in a city like Ahmedabad that predicts traffic congestion using both transportation and climate variables.

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https://www.ahmedabad ↗ mirror.com/traffic-woes-at-30-major-junctions/81853476.html

Easing 'pain in the neck' SVNIT to study 30 junctions in the city with highest traffic congestion, suggest solutions

---

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Imagine you are designing an intelligent system in a city like Ahmedabad that predicts traffic congestion using both transportation and climate variables.

Step 1: Define N Random Variables

Let's define a vector of random variables:

This is your N-dimensional random vector (here $N = 5$).

Step 2: Mean Vector

Case Study Continued

Step 3: Covariance Matrix Interpretation (Intuition and By Definition)

- More rain ⇒ more congestion (traffic increases)
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

covariance matrix = "dependency map of the city"

Step 4: Joint Gaussian Distribution

We assume:

This means:

The entire system (traffic + climate) behaves in a structured probabilistic way defined by:

- Mean (typical conditions)
- Covariance (interdependencies)

Why Gaussian?

Step 6: Transformation

Now suppose the Ahmedabad city defines a Congestion Index:

This is a linear transformation:

Inference # 1

Step 6: Transformation (Continued)

Now suppose the Ahmedabad city defines a Congestion Index:

This is a linear transformation:

If rainfall increases suddenly, what happens?

- Speed ↓
- Traffic ↑
- Pollution ↑

Inference # 2

Repeated Inference Slide

If rainfall increases suddenly, what happens?

- Speed ↓
- Traffic ↑
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Inference # 2

📄 FULL LATEX CODE (COPY-PASTE READY)

LaTeX

```
\documentclass[12pt]{article}
\usepackage{amsmath}
\usepackage{geometry}
\usepackage{graphicx}
\geometry{margin=1in}

\title{CSE400 - Fundamentals of Probability in Computing\\
Lecture 20: Linear Transformations and Expectations of Multiple RVs}
\author{Dhaval Patel, PhD}
\date{March 19, 2025}

\begin{document}

\maketitle

\section*{}
```

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\item Traffic $\uparrow$
\item Pollution $\uparrow$
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\textbf{Inference \# 2}

\subsection*{Repeated Inference Slide}

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\end{document}
```

⚠️ Important Notes (honest and necessary)

- Your uploaded PDF is **mostly slide headings + visuals**, so there are:
  - **No explicit formulas**
  - **No full derivations shown in text**
- I have **NOT added anything extra** (as you requested)
- I have **preserved wording exactly** wherever visible text existed

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If you want, I can next:

- Convert this into a **proper exam-ready scribe with formulas (still based only on your slides)**
- Or generate **expected viva questions from this lecture**

Just tell me 👍